

Charlie Marx

CURRICULUM VITAE - DECEMBER 2019

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EDUCATION

Haverford College

B.S. Candidate in Computer Science

B.S. Candidate in Mathematics

Minor Candidate in Statistics

2016 – PRESENT

GPA: 3.94/4.0

Relevant Coursework

Machine Learning · Advanced Topics in Probability · Real Analysis · Abstract Algebra

Multivariable Calculus · Programming Languages · Linear Algebra · Analysis of Algorithms

AWARDS & HONORS

Beckman Scholars Award

2018

National award for excellence in research including \$26,000 of research and travel funding.

Brian Kovaric Award Finalist

2017

Faculty-nominated scholarship awarded to one outstanding first-year student.

Farmers Insurance Standout Student of the Year

2016

Regional award for academic and community engagement awarded to one student each year.

PUBLICATIONS

MACHINE LEARNING METHODS

1. [Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)

Charles T. Marx, Richard L. Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian

NeurIPS – Conference on Neural Information Processing Systems, 2019

2. [On the Multiplicity of Predictions in Classification](#)

Charles T. Marx, Flavio du Pin Calmon, Berk Ustun

NeurIPS Workshop on Human-Centered Machine Learning, 2019

Accepted for Oral Presentation

3. [Predictive Multiplicity in Classification](#)

Charles T. Marx, Flavio du Pin Calmon, Berk Ustun

In Submission

MACHINE LEARNING APPLICATIONS

4. [Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill](#)

Helen K. White, **Charles T. Marx**, David L. Valentine, Charles Sharpless, Christoph Aeppli, Kelsey M. Gosselin, Veronika Kivenson, Rachel M. Liu, Robert K. Nelson, Sean P. Sylva, Christopher M. Reddy

ACS Earth and Space Chemistry, 2019

5. [Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence](#)

Anna P. M. Michel, Alexandra E. Morrison, **Charles T. Marx**, Helen K. White

ACS Environmental Science & Technology Letters, 2018

6. [Comparison of Laboratory Analytical Techniques for the Rapid Identification of Marine Plastics](#)

Anna P. M. Michel, Alexandra Morrison, Beckett Colson, **Charles T. Marx**, Helen K. White

OCEANS, 2020

RESEARCH EXPERIENCE

Research Assistant in Machine Learning *Harvard University*

2019

Identified new measures for multiplicity in machine learning. Developed software, proved theorems bounding multiplicity, and managed experiments on a research computing cluster.

Advisors: Berk Ustun and Flavio Calmon

Research Assistant in Machine Learning *Haverford College*

2017 – PRESENT

Designed new methods to quantify indirect feature influence in machine learning models using disentangled representations and adversarial learning.

Advisor: Sorelle Friedler

	Research Assistant in Chemistry <i>Haverford College</i>	2017 – 2018
	Designed machine learning tools to optimize model performance, robustness, and interpretability for on-site oil identification.	
	<i>Advisor:</i> Helen White	
	Summer Research Intern <i>Prince William Sound Science Center</i>	2018
	Automated salmon and herring tracking in Alaska using machine learning and sonar data.	
	<i>Advisor:</i> Scott Pegau	
	Research Assistant in Computer Vision <i>Haverford College</i>	2017
	Analyzed 100,000 handwritten texts using recurrent and convolutional neural networks.	
	<i>Advisor:</i> Andrew Janco	
ORAL PRESENTATIONS	UC Berkeley Simons Institute for the Theory of Computing <i>Berkeley, CA</i>	2019
	Disentangling Influence (45 minute talk)	
	INFORMS Annual Meeting <i>Seattle, WA</i>	2019
	Predictive Multiplicity in Classification (20 minute talk)	
	FAT* Conference <i>New York, NY</i>	2018
	Auditing Black Box Models (60 minute tutorial)	
	<i>with Carlos Scheidegger and Suresh Venkatasubramanian</i>	
	GOMOSES Conference <i>New Orleans, LA</i>	2018
	Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15 minute talk)	
POSTER PRESENTATIONS	NeurIPS Human-Centered Machine Learning Workshop <i>Vancouver, BC</i>	2019
	On the Multiplicity of Predictions in Machine Learning	
	Beckman National Symposium <i>Irvine, CA</i>	2019
	Indirect Influence in Machine Learning Models	
	Beckman Regional Symposium <i>New York, NY</i>	2018
	Mitigating Social Discrimination in Deep Learning	
	KINSC Undergraduate Research Symposium <i>Haverford, PA</i>	2017
	Examining Biogenic Hydrocarbons After the Deepwater Horizon Oil Spill	
CONFERENCE PARTICIPATION	NeurIPS <i>Conference on Neural Information Processing Systems</i>	2019
	INFORMS <i>Institute for Operations Research and the Management Sciences</i>	2019
	FAT* <i>Fairness, Accountability and Transparency in ML</i>	2018, 2019
	KDD <i>International Conference on Knowledge Discovery & Data Mining</i>	2018
	GOMOSES <i>Gulf of Mexico Oil Spill & Ecosystem Science Conference</i>	2017
WORKSHOP PARTICIPATION	Summer Cluster on Fairness at the UC Berkeley Simons Institute	2019
	Workshop on Functional Ecological Genomics	2017
FIELDWORK	Alaska Prince William Sound Research Charter	2018
	Gulf of Mexico Shoreline Research Tour	2017
ACADEMIC ACTIVITIES	Teaching Assistant in Machine Learning	2019 – PRESENT
	Held office hours and graded for an upper-level machine learning course.	
	Peer Tutor	2018 – 2019
	Tutored students in statistics, linear algebra, and machine learning.	
	Computer Science Faculty Search Committee	2018 – 2019
	Student representative on computer science tenure-track faculty search committee.	
EXTRA- CURRICULAR ACTIVITIES	Captain of Haverford College Men's Club Soccer	2018 – 2019
	Leader of Haverford College Outdoors Club	2017 – 2018
	President of Furniture for Newly Housed Families Organization	2015 – 2017